Hospital Engineering and Management

Lecture 15: Centralized Medical Gas and Vacuum Systems

Dr. Sahaj Saxena

Department of Electrical and Instrumentation Engineering TIET

Course Code: UBM702

Learning Objectives

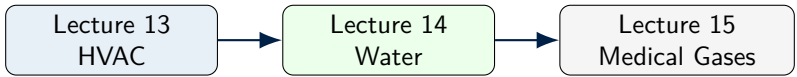
By the end of this lecture, students should be able to:

- identify the major medical gases, vacuum and anaesthetic-gas disposal services used in hospitals;
- explain the complete medical-gas pipeline system from source plant to terminal unit;
- distinguish cylinder manifolds, liquid-oxygen systems and oxygen-generation plants;
- describe compressors, vacuum pumps, dryers, filters, regulators, valves and alarm systems;
- perform preliminary demand, reserve and pipeline-flow calculations;
- recognize hazards related to pressure, oxygen enrichment, cryogenic liquids, contamination and cross-connections;
- relate MGPS design, commissioning and maintenance to biomedical equipment and patient safety.

Connection with Lecture 14

Lecture 14 Established

Hospital water, plumbing and sanitation systems are engineered utilities whose failure can rapidly affect clinical care.



Key Transition

Medical-gas systems are different from ordinary building utilities because the delivered product can directly sustain life.

Opening Scenario: Oxygen Pressure Falls in an ICU

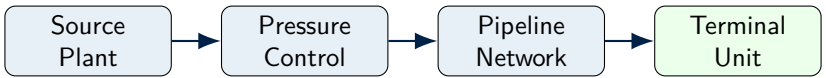
At 2:00 a.m., the oxygen pipeline pressure in a 20-bed ICU begins to fall while several patients are mechanically ventilated.

- How will staff know that source pressure is falling?
- What reserve source should automatically become available?
- Which valves should remain open and which can be isolated?
- Can portable cylinders sustain all ventilated patients?
- Who coordinates the clinical and engineering response?

Medical-gas engineering is directly connected to continuity of life support.

What Is a Medical Gas Pipeline System?

A medical gas pipeline system (MGPS) is a centralized infrastructure that produces or stores gases and vacuum, conditions them, distributes them through dedicated pipelines, and delivers them safely at clinical terminal units.



Core Requirement

Correct gas + correct pressure + adequate flow + uninterrupted availability + no contamination.

Why Centralized Medical Gases?

Centralized systems can provide:

- continuous bedside availability;
- reduced movement of cylinders through patient areas;
- controlled and regulated delivery pressure;
- centralized alarms and source monitoring;
- improved workflow in ICU, OT and emergency areas;
- easier capacity planning for high-acuity services;
- safer segregation of bulk gas storage from occupied clinical spaces.

Important

Centralization increases convenience, but it also creates the possibility of a common-mode failure affecting many patients simultaneously.

Major Services in an MGPS

Positive-Pressure Services

- oxygen (O_2);
- medical air;
- nitrous oxide (N_2O);
- carbon dioxide (CO_2), where clinically required;
- nitrogen / surgical-tool air, where applicable.

Negative / Disposal Services

- medical vacuum;
- anaesthetic gas scavenging/disposal systems;
- dedicated specialty suction systems where required.

Medical Oxygen

Medical oxygen is the most safety-critical gas in most hospitals.

- supports oxygen therapy, ventilation and anaesthesia;
- may be supplied by cylinders, liquid oxygen or an on-site generation system;
- is non-flammable but strongly supports combustion;
- requires dedicated clean pipework and components compatible with oxygen service;
- must remain available during source, power or distribution failures.

Clinical consequence of failure: hypoxaemia, interruption of ventilation and possible loss of life.

Medical Air

Medical air is compressed air prepared for patient-care use.

- used in ventilators, anaesthesia systems and respiratory therapy;
- generated by medical-air compressor plants or supplied from approved sources;
- requires filtration, drying and removal of contaminants;
- should not be confused with ordinary workshop or industrial compressed air;
- quality depends on compressor design, intake location and treatment stages.

Engineering Principle

A compressor provides pressure; the treatment train provides medical-air quality.

Nitrous Oxide

Nitrous oxide may be used as an anaesthetic/analgesic gas.

- usually supplied from cylinders or manifolds;
- must remain completely separated from oxygen and air pipelines;
- leakage contributes to occupational exposure and environmental emissions;
- terminal identification and pipeline verification are essential;
- anaesthetic-gas scavenging becomes important where inhalational agents are used.

Safety

A wrong-gas connection is a potentially catastrophic latent engineering error.

Other Clinical Gases

Depending on hospital services, specialized systems may include:

- carbon dioxide for minimally invasive surgery/endoscopy;
- nitrogen for selected surgical tools or laboratory applications;
- high-pressure surgical/tool air;
- specialty gas mixtures for diagnostics or therapy.

Design Rule

Do not install a gas simply because it is available. The clinical need, source, storage, distribution, terminal type, alarms and maintenance capability must all be justified.

Medical Vacuum

Medical vacuum provides controlled suction at the point of care.

- airway and oral suction;
- surgical suction;
- drainage and selected clinical procedures;
- collection through appropriate suction regulators and containers.

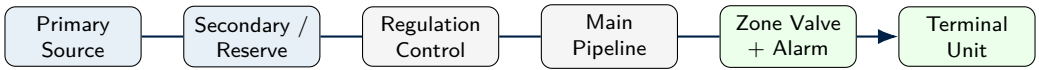


Anaesthetic Gas Scavenging System

Anaesthetic gas scavenging reduces staff exposure to waste anaesthetic gases.

- collects excess gases from anaesthesia breathing systems;
- transfers them through a dedicated disposal pathway;
- must not impose unsafe positive or negative pressure on the patient circuit;
- is distinct from ordinary medical vacuum unless specifically designed otherwise;
- requires maintenance and functional verification.

Complete MGPS Architecture



Safety Layers

Source redundancy, pressure regulation, isolation valves, alarms, labelled dedicated pipelines and terminal-specific connectors form multiple independent barriers.

Source-System Philosophy

A source system should be designed so that loss of one component does not immediately interrupt clinical supply.

- primary source: normal operating source;
- secondary source: automatically or rapidly available alternate source;
- reserve source: supports supply when normal sources are unavailable or depleted;
- source alarms: indicate low pressure, changeover and plant faults;
- sufficient stored/produced capacity for demand and emergency planning.

Cylinder Manifold Systems

Cylinder manifolds connect banks of cylinders to a common regulated supply.

- two-bank arrangement permits duty/standby operation;
- automatic changeover can maintain supply when one bank is depleted;
- non-return valves help prevent reverse flow;
- regulators reduce cylinder pressure to pipeline pressure;
- cylinders require secure storage, ventilation and disciplined inventory control.

$$\text{Duty Bank} \rightarrow \text{Regulator} \rightarrow \text{Pipeline} \leftarrow \text{Regulator} \leftarrow \text{Standby Bank}$$

Bulk Liquid Oxygen System

Large hospitals may use vacuum-insulated evaporator systems for bulk liquid oxygen.

- oxygen is stored cryogenically as a liquid;
- vaporizers convert liquid oxygen to gas before distribution;
- large storage density reduces the number of cylinders handled;
- pressure, level, relief devices and reserve supply must be monitored;
- siting must consider tanker access, separation, ventilation and emergency response.

Cryogenic Hazard

Extremely low temperatures can cause cold burns and material embrittlement; released oxygen can create an oxygen-enriched atmosphere.

On-Site Oxygen Generation: PSA

Pressure Swing Adsorption (PSA) plants can generate oxygen-enriched product gas on site.

Air $\rightarrow$ Compression $\rightarrow$ Drying/Filtration $\rightarrow$ Molecular Sieve $\rightarrow$ Product Receiver $\rightarrow$ Pipeline

- reduces dependence on delivered cylinders/liquid oxygen;
- requires reliable power and compressor redundancy;
- product quality and concentration must be continuously controlled;
- backup oxygen storage remains essential;
- maintenance capability becomes part of clinical resilience.

Medical Air Compressor Plant

A typical plant includes:

- multiple compressors for redundancy;
- air intake located away from contamination sources;
- aftercoolers and moisture separators;
- dryers;
- particulate and coalescing filters;
- receiver tank;
- pressure regulation and alarms.

Key Question

If one compressor fails during peak demand, can the remaining plant still maintain required flow and pressure?

Medical Vacuum Plant

A centralized vacuum plant commonly contains:

- multiple vacuum pumps;
- receiver(s);
- bacterial/particulate filtration where required by system design;
- isolation valves;
- controls and alarms;
- safe exhaust arrangement.

Reliability

Duty cycling can distribute operating hours across pumps and improve maintainability.

Receivers, Dryers and Filters

Ancillary components strongly influence gas quality and plant stability.

- receivers damp rapid demand fluctuations;
- dryers reduce moisture that can damage equipment or degrade gas quality;
- filters remove particles, oil aerosols and other contaminants;
- differential pressure indicates filter loading;
- drains must be managed without contaminating the system.

Engineering Lesson

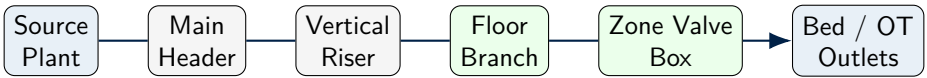
A compressor plant is not complete until air quality, moisture control and filtration are verified.

Pipeline Materials and Installation

Medical-gas pipelines require controlled material selection and workmanship.

- tubing/components must be suitable for the intended gas service;
- pipe interiors must remain clean, dry and free from oil/grease contamination;
- joints should be made using approved procedures;
- pipelines must be identified and protected from accidental damage;
- branch layouts should support isolation and future maintenance;
- installation cleanliness is verified before clinical use.

Pipeline Network: Conceptual Layout



- larger mains carry aggregated flow;
- branch pipes serve defined clinical zones;
- isolation should not unnecessarily interrupt unrelated departments.

Pipeline Sizing: Basic Engineering Idea

For preliminary analysis:

$$Q = Av$$

where Q is volumetric flow, A is pipe cross-sectional area and v is average gas velocity. The selected diameter must also control:

- pressure drop from source to the most remote outlet;
- expected simultaneous demand;
- future expansion allowance;
- noise and velocity;
- gas-specific design requirements.

Demand Is Not Simply Outlet Count

If a ward has 40 oxygen outlets, it does not necessarily mean all 40 operate at maximum flow simultaneously.

- design demand depends on clinical area and patient acuity;
- diversity/simultaneity assumptions may be applied according to accepted design practice;
- ICU and OT have different demand patterns from general wards;
- emergency surge conditions may exceed normal demand;
- future beds and equipment should be considered.

Do Not Guess

Final medical-gas sizing must follow the applicable standard, clinical brief and authority requirements.

Numerical Example: Pipe Velocity

Suppose a branch carries $Q = 600$ L/min of gas through a pipe with internal diameter 25 mm.

$$Q = 600 \times 10^{-3}/60 = 0.010 \text{ m}^3/\text{s}$$

$$A = \frac{\pi D^2}{4} = \frac{\pi(0.025)^2}{4} = 4.91 \times 10^{-4} \text{ m}^2$$

$$v = \frac{Q}{A} = \frac{0.010}{4.91 \times 10^{-4}} \approx 20.4 \text{ m/s}$$

Interpretation

The calculation gives only velocity. A real design must also verify pressure drop, operating pressure, demand model and code limits.

Terminal Units

Terminal units are the clinical connection points between the fixed pipeline and equipment.

- should be gas-specific and clearly identified;
- designed to reduce accidental interchange;
- may be wall-mounted, bed-head mounted, pendant-mounted or ceiling-mounted;
- require adequate number and location for clinical workflow;
- should permit maintenance and leak testing;
- connection integrity must be checked during commissioning.

Zone Valve Boxes

Zone valves allow a defined clinical area to be isolated without shutting down the entire hospital.

- installed at accessible, identifiable locations;
- serve specific areas such as ICU, OT suite or ward;
- should not be hidden behind furniture or locked without controlled access;
- labelling should state exactly which area is isolated;
- valve closure during an emergency must be coordinated with clinical staff.

Isolation is a safety function only when the correct zone can be identified quickly.

Area Alarm Panels

Area alarms monitor the local pressure/vacuum condition of a clinical zone.

- high/low oxygen pressure;
- high/low medical-air pressure;
- loss of vacuum performance;
- other gas-specific alarm conditions;
- visual and audible annunciation.

Human Factors

An alarm that is technically correct but poorly located, poorly labelled or routinely ignored does not provide effective protection.

Master Alarms

Master alarm systems communicate plant/source status to engineering and other responsible locations.

- source changeover;
- low reserve content;
- compressor/pump fault;
- high or low pipeline pressure;
- power or control fault;
- PSA oxygen-quality alarm, where applicable.

Alarm Philosophy

Critical alarms must be actionable: they should identify what failed, where it failed and who must respond.

Pressure Regulation

Pressure regulators reduce and stabilize gas pressure before distribution or at selected stages.

- prevent high source pressure reaching the clinical network;
- maintain pressure within the intended operating band;
- often arranged with redundancy/bypass according to design;
- should be sized for expected peak flow;
- require periodic inspection and functional testing.

Identification, Labelling and Colour Coding

Safe identification uses multiple independent cues.

- gas name/chemical symbol;
- pipeline labels and direction markings;
- gas-specific terminal connectors;
- cylinder identification according to applicable standards;
- valve-zone labels;
- drawings and asset documentation.

Important

Never rely on colour alone to identify a medical gas. Read the label and verify the connection.

MGPS in the Operation Theatre

An OT may require multiple medical-gas services at pendants or wall terminals.

- oxygen;
- medical air;
- nitrous oxide, where used;
- vacuum;
- anaesthetic-gas scavenging;
- CO_2 for minimally invasive procedures, where centrally supplied.

Planning Challenge

Terminal location must support anaesthesia and surgery without creating hose entanglement, trip hazards or obstruction of equipment movement.

MGPS in ICU and High-Dependency Areas

Critical-care beds have high and variable utility demand.

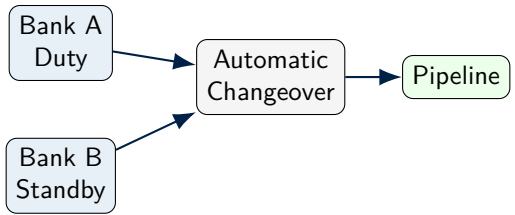
- multiple oxygen outlets may be required per bed;
- medical air supports ventilators and respiratory devices;
- vacuum supports suction procedures;
- bed-head units/pendants integrate gas, power and data;
- terminal accessibility must be preserved behind equipment;
- emergency cylinders should be planned as part of contingency procedures.

MGPS in Emergency and Resuscitation Areas

Emergency departments need rapid access under unpredictable peaks.

- resuscitation bays require reliable oxygen and suction;
- mass-casualty events may greatly increase oxygen demand;
- portable cylinders may supplement fixed systems;
- reserve capacity should reflect disaster planning;
- hose routing and outlet access must permit rapid patient turnover.

Automatic Manifold Changeover



- low pressure/depletion on duty bank initiates transfer;
- alarm indicates that reserve bank is now in use;
- depleted cylinders must be replaced before the remaining bank is exhausted.

Reserve Supply Is Not Optional Thinking

A resilient design asks: “What if the primary source is unavailable?”

- tanker delay or bulk source depletion;
- PSA plant electrical/compressor failure;
- manifold regulator fault;
- maintenance shutdown;
- pipeline damage;
- disaster surge in demand.

Reserve capacity should be linked to credible failure scenarios and replenishment time.

Redundancy: N+1 Concept

If N components are required to meet peak demand, an N+1 configuration provides one additional component.

$$\text{Installed components} = N + 1$$

Example: if two compressors are required to meet peak medical-air demand, three compressors provide one-unit redundancy.

Caution

N+1 is useful only if common dependencies such as electrical supply, cooling, controls or headers do not defeat the redundancy.

Preliminary Source Capacity Calculation

A simplified planning estimate can be written as:

$$Q_{design} = \sum_{i} n_i q_i f_i$$

where:

- n_i = number of outlets/beds in clinical category i;
- q_i = assumed unit flow;
- f_i = simultaneity/diversity factor;
- Q_{design} = estimated design flow.

Use Carefully

Values of q_i and f_i must come from the applicable design standard and clinical brief, not from arbitrary assumptions.

Numerical Example: Oxygen Demand

For a teaching example, suppose:

- 20 ICU beds at 15 L/min with simultaneity 0.8;
- 60 ward beds at 5 L/min with simultaneity 0.3.

Then:

$$Q_{ICU} = 20 \times 15 \times 0.8 = 240 \text{ L/min}$$

$$Q_{ward} = 60 \times 5 \times 0.3 = 90 \text{ L/min}$$

$$Q_{total} = 330 \text{ L/min}$$

This is a teaching estimate only; actual design values must follow the governing standard and clinical risk assessment.

Numerical Example: Emergency Oxygen Reserve

If emergency demand is estimated at 300 L/min and the target reserve duration is 4 hours:

$$V = 300 \times 60 \times 4 = 72{,}000 \text{ L}$$

Thus the reserve must provide at least 72,000 L of usable oxygen at the required delivery conditions, before adding operational margin and considering cylinder residual pressure.

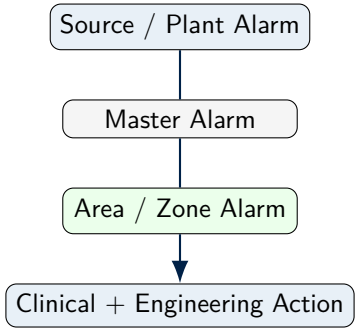
Engineering Question

How many cylinders are needed? The answer depends on cylinder water capacity, filling pressure, usable residual pressure and applicable safety practice.

Choosing the Oxygen Source

	Cylinder Manifold	Liquid Oxygen	PSA Plant
Best fit	Low/moderate demand	High continuous demand	Continuous on-sit generation
Dependency	Cylinder logistics	Bulk refill/tanker	Power + compressors
Space	Cylinder store	Cryogenic compound	Plant room
Backup	Additional banks	Cylinder/alternate source	Cylinder/alternate source
Key risk	Depletion/changeover	Cryogenic + O_2 enrichment	Quality/power failure

Alarm Hierarchy



Principle

Different alarms answer different questions: Is the plant healthy? Is the hospital network healthy? Is this clinical zone safe?

Commissioning: Why It Is Critical

An MGPS should never be assumed safe simply because installation is complete.

- verify pipeline identity and continuity;
- pressure/leak testing;
- verify terminal-unit gas identity;
- confirm flow and pressure performance;
- verify alarms and valve-zone labels;
- check source changeover and reserve operation;
- document results before clinical handover.

Cross-Connection Testing

A cross-connection occurs when one gas pipeline is incorrectly connected to another service.

Why It Is Dangerous

The pipeline may appear pressurized and operational while delivering the wrong gas to a patient.

Controls include:

- independent verification of gas identity at every terminal;
- gas-specific connectors;
- controlled labelling;
- commissioning tests after modifications;
- strict change-control documentation.

Contamination Risks

Medical gases can be compromised by:

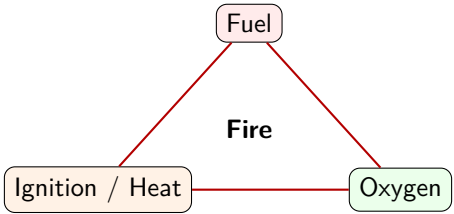
- oil/grease in oxygen-service components;
- moisture in compressed-air pipelines;
- particles introduced during construction;
- contaminated compressor intake air;
- incorrect cleaning products;
- corrosion or construction debris;
- wrong source connection.

Control

Clean construction, purging, testing, filtration and documented commissioning are as important as mechanical installation.

Oxygen Enrichment and Fire Risk

Oxygen does not itself burn, but increased oxygen concentration makes many materials ignite more easily and burn more intensely.



Implication

Smoking, flames, oil/grease contamination and uncontrolled oxygen leakage must be prevented around oxygen systems.

Oxygen Fire-Safety Controls

- control ignition sources;
- use oxygen-compatible components and cleaning practices;
- secure and ventilate storage/source areas;
- repair leaks promptly;
- keep combustible materials away from bulk/manifold areas;
- provide clear emergency isolation procedures;
- train clinical and engineering staff in oxygen hazards.

Cylinder Storage and Handling

Compressed-gas cylinders store substantial mechanical energy.

- keep cylinders secured against falling;
- protect valve assemblies from damage;
- separate full and empty cylinders operationally;
- maintain clear identification;
- use appropriate trolleys for transport;
- avoid heat exposure and unauthorized adapters;
- ensure storage ventilation and access control.

Cryogenic Oxygen Safety

Liquid oxygen introduces additional hazards:

- extreme cold and frost injury;
- rapid expansion from liquid to gas;
- oxygen enrichment after release;
- pressure buildup if liquid is trapped;
- tanker movement and transfer operations.

Engineering Controls

Relief devices, safe siting, ventilation, restricted access, signage and trained transfer procedures are essential.

Leak Detection and Response

Possible indicators of gas leakage include:

- unexpected source depletion;
- abnormal compressor/pump running time;
- pressure instability;
- audible leakage at terminal units or valves;
- unexplained oxygen-enrichment alarms where installed.

Response

Locate the leak, assess clinical dependence, isolate only when safe, provide alternate supply, repair, retest and document.

Preventive Maintenance of MGPS

Maintenance should include planned checks of:

- compressors and vacuum pumps;
- manifold regulators and changeover units;
- dryers and filters;
- receivers and relief devices;
- alarm panels and sensors;
- zone valves and terminal units;
- source levels and reserve status;
- electrical/control systems.

Condition Monitoring

Useful condition indicators include:

- compressor running hours and starts/hour;
- discharge temperature;
- filter differential pressure;
- dryer dew point;
- vacuum level;
- pipeline pressure trends;
- bulk oxygen level;
- PSA oxygen concentration and purity-related alarms.

Planned Shutdown of a Medical Gas Zone

Before closing a valve for maintenance:

1. identify exactly which outlets are affected;
2. obtain clinical approval and communicate timing;
3. identify every dependent patient/device;
4. arrange alternative supply;
5. isolate and perform work;
6. test pressure, gas identity and terminal function;
7. obtain clinical confirmation before return to service.

Never

Do not close a medical-gas zone valve based only on a drawing without confirming the actual served area.

Emergency Oxygen Failure Response

A hospital emergency procedure should define:

- how the failure is detected and escalated;
- who assesses source vs distribution failure;
- how portable cylinders are deployed;
- priority patients and clinical areas;
- how supply is conserved safely;
- how external replenishment is requested;
- when evacuation or service restriction is considered.

Case Study 1: Primary Oxygen Source Depleted

A hospital's primary oxygen source reaches a critical low level earlier than expected.

Possible Causes

Demand surge, leak, delayed delivery, faulty level indication or planning error.

1. Confirm actual source level and current consumption.
2. Verify secondary/reserve readiness.
3. Estimate time-to-depletion.
4. Notify clinical leadership and supplier.
5. Search for abnormal leakage or demand.

Engineering lesson: "remaining hours" is often more useful than "percent full."

Case Study 2: Pressure Drop in a Remote ICU

The plant pressure is normal, but the farthest ICU reports low oxygen pressure during peak demand.

- undersized branch or main pipeline;
- partially closed valve;
- obstructed regulator/filter;
- excessive unplanned connected load;
- local leak;
- inaccurate pressure sensing.

Diagnostic Approach

Compare pressure sequentially from plant $\rightarrow$ main $\rightarrow$ riser $\rightarrow$ zone $\rightarrow$ terminal while recording flow demand.

Case Study 3: Wrong-Gas Connection after Renovation

After renovation, a newly commissioned outlet is suspected of delivering the wrong gas.

Immediate Priority

Stop clinical use of the affected outlet/zone and provide an independently verified alternate supply.

Then:

- quarantine affected terminals;
- trace pipeline connections;
- test gas identity at all modified outlets;
- review commissioning records;
- perform root-cause analysis before reopening.

Case Study 4: Alarm Fatigue

A plant alarm sounds frequently because of a recurring non-critical sensor fault. Staff begin silencing it without investigation.

- What happens when a genuine source failure occurs?
- Is the problem technical, organizational or both?
- Should an alarm ever be permanently muted?

Lesson

Repeated nuisance alarms degrade trust in the entire safety system. Alarm defects require root-cause correction, not normalization.

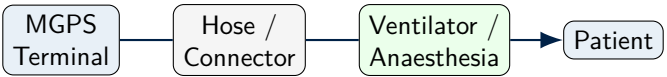
Biomedical Engineer's Role in MGPS

Biomedical engineers may contribute to:

- clinical requirement definition during planning;
- interface review between pipelines and medical devices;
- terminal location and equipment compatibility;
- commissioning support and acceptance testing;
- alarm escalation and incident investigation;
- preventive maintenance coordination;
- contingency planning for ventilators and anaesthesia systems.

MGPS–Medical Device Interface

Pipeline gas is only one part of the clinical delivery chain.



A failure can occur at any interface: outlet pressure, connector, hose, device regulator, internal valve or patient circuit.

Effect on Mechanical Ventilation

Many ventilators depend on wall oxygen and/or compressed air.

- low inlet pressure may trigger device alarms;
- some ventilators can continue temporarily using alternate pneumatic/electrical architectures;
- oxygen concentration may be affected by source availability;
- portable cylinder duration must be calculated before transport or outage;
- engineering and respiratory/critical-care teams must understand device-specific behavior.

Anaesthesia Workstation Interface

Anaesthesia workstations may connect to oxygen, medical air, nitrous oxide and scavenging systems.

- pipeline pressure is reduced internally by device regulators;
- gas-specific inlets reduce interchange risk;
- backup cylinders may provide local redundancy;
- low oxygen pressure alarms are safety-critical;
- scavenging function protects staff and environment.

Clinical Engineering Task

Acceptance testing should verify both the device and its interfaces with the hospital utility system.

Flowmeters, Regulators and Suction Controllers

Bedside accessories convert the fixed utility into a clinically usable service.

- oxygen flowmeters regulate patient flow;
- pressure regulators control cylinder delivery pressure;
- vacuum regulators limit suction pressure;
- collection jars/traps protect the vacuum system;
- accessories require cleaning, inspection and functional checks.

Infection-Control Considerations

Medical-gas and vacuum maintenance can affect infection control.

- contaminated suction containers should not expose the fixed vacuum network;
- terminal units should be maintained without introducing debris;
- work in clinical zones may require dust-control measures;
- filters and drains require safe handling;
- pipeline modifications require controlled cleaning and commissioning.

Monitoring through BMS / Digital Platforms

Useful remotely monitored parameters may include:

- source pressure and level;
- manifold bank status;
- compressor/pump running state;
- plant fault alarms;
- pipeline pressures;
- dew point/quality parameters;
- PSA oxygen quality;
- alarm acknowledgement and response time.

Caution

Digital monitoring supplements local safety alarms; it should not become the only protection layer.

Useful MGPS Performance Indicators

Examples include:

- number of unplanned gas-supply interruptions;
- alarm response time;
- reserve-source availability;
- compressor/vacuum-pump downtime;
- preventive-maintenance compliance;
- oxygen consumption per occupied bed-day;
- leakage-related consumption trend;
- percentage of terminal units passing periodic functional checks.

FMEA Example: Oxygen Supply

Failure mode	Effect	Control
Primary source empty	Falling pressure	Secondary/reserve + low-level alarm
Regulator failure	High/low pressure	Redundant regulation + alarms
Pipeline leak	Capacity loss	Pressure monitoring + leak testing
Wrong connection	Wrong gas to patient	Gas-specific design + independent commissioning
Zone valve closed	Local outage	Labelling + controlled shutdown procedure
Alarm ignored	Delayed response	Escalation + alarm governance

Single-Point-of-Failure Analysis

Ask whether one event can interrupt the entire service.

- one electrical feeder for all compressors;
- one common controller for redundant pumps;
- one isolation valve on the only hospital oxygen main;
- reserve cylinders stored but not connected;
- alarms visible only inside an unattended plant room.

Redundant components do not guarantee a redundant system.

Energy and Sustainability Considerations

Medical-gas systems also have resource and energy implications.

- compressors and vacuum pumps consume significant electricity;
- compressed-air leaks waste energy;
- oxygen leakage wastes a clinical resource;
- efficient sequencing reduces unnecessary compressor operation;
- heat recovery may be possible from compressor systems;
- nitrous oxide and anaesthetic-gas emissions have environmental implications.

Design Review Checklist

Before approving an MGPS design, ask:

1. Are clinical gases and outlet counts justified?
2. Is peak demand calculated?
3. Is source redundancy adequate?
4. Can each clinical zone be isolated safely?
5. Are alarms located where staff will respond?
6. Are routes protected and maintainable?
7. Is expansion possible?
8. Is commissioning responsibility clearly assigned?

Classroom Activity 1: Design an ICU Gas System

A new 12-bed ICU will have ventilated patients and two isolation rooms.

1. Which medical-gas/vacuum services should be considered at each bed?
2. Where should the area alarm be located?
3. How would you define the zone-valve boundary?
4. What emergency oxygen backup would you plan?
5. Which parameters should be monitored by engineering?

Work in groups for 7 minutes and draw a block diagram.

Classroom Activity 2: Reserve Calculation

A critical-care area is estimated to consume 240 L/min of oxygen during an outage.

1. Estimate oxygen volume required for 3 hours.
2. Add a 20% planning margin.
3. List three reasons actual required reserve could be larger.

$$V = Q \times t$$

Classroom Activity 2: Quick Solution

For 3 hours:

$$V = 240 \times 60 \times 3 = 43{,}200 \text{ L}$$

With a 20% planning margin:

$$V_{design} = 43{,}200 \times 1.20 = 51{,}840 \text{ L}$$

Possible reasons for additional reserve:

- surge in patient demand;
- usable cylinder volume is less than nominal theoretical content;
- refill/replenishment delay;
- leakage or unexpected additional connected loads.

Classroom Activity 3: Find the Hidden Risk

Identify the dominant risk in each situation:

1. A zone valve has no label indicating which ward it serves.
2. Oxygen cylinders are stored unsecured near a heat source.
3. A medical-air compressor intake is near a vehicle exhaust area.
4. A renovated outlet was not independently gas-identity tested.
5. A plant alarm has been muted for weeks because it is "always false."

Mini-Project Ideas

- IoT dashboard for oxygen-source level and consumption forecasting.
- Medical-gas alarm prioritization and escalation prototype.
- ICU oxygen reserve calculator for emergency planning.
- FMEA of an MGPS in a 300-bed hospital.
- Leak-detection strategy using pressure/flow trend analytics.
- Comparative techno-economic study: cylinders vs LOX vs PSA oxygen.
- Digital preventive-maintenance schedule for MGPS components.

Standards and Guidance Framework

Medical-gas design and operation should follow the edition adopted by the local authority and project specification. Common reference frameworks include:

- ISO 7396-1, medical gas pipeline systems – compressed medical gases and vacuum;
- NFPA 99, Health Care Facilities Code, where contractually/regulatorily applicable;
- national building, fire, pressure-vessel and gas-storage requirements;
- hospital accreditation requirements related to utility safety and continuity;
- manufacturer requirements for connected medical devices.

Teaching Note

Standards define detailed design values and test procedures. This lecture focuses on engineering principles and system thinking.

What Biomedical Students Should Remember

- A gas outlet is the end of a complete life-support utility chain.
- Source redundancy and reserve planning matter as much as normal operation.
- Pressure, flow, gas identity and purity/quality are different safety dimensions.
- Zone valves and alarms convert a large network into manageable safety zones.
- Commissioning prevents hidden cross-connections and installation errors.
- Medical devices and MGPS must be considered as one integrated clinical system.

Summary

- MGPS delivers oxygen, medical air, specialty gases, vacuum and disposal services safely to clinical areas.
- Source plants, regulators, pipelines, valves, alarms and terminal units form a layered safety system.
- Reliability depends on redundancy, reserve capacity, monitoring and maintenance.
- Wrong-gas connections, contamination, pressure loss, leakage and oxygen enrichment are major hazards.
- Commissioning and controlled shutdown procedures are essential patient-safety activities.
- Biomedical engineers bridge utility infrastructure and the medical devices that depend on it.

Medical gases are utilities with clinical consequences.

Assignment

Prepare a two-page engineering note for the MGPS of a **300-bed multi-specialty hospital**.

1. Draw the oxygen system from primary source to patient terminal.
2. Show primary, secondary and reserve supply philosophy.
3. Identify ICU, OT and ward zone-valve boundaries.
4. Propose five critical alarms and their response locations.
5. Identify five credible failure modes and controls.
6. Perform one numerical estimate of oxygen demand or emergency reserve.

Include

One block diagram, one small calculation and one FMEA-style table.

Suggested Reading

- ISO 7396-1, *Medical Gas Pipeline Systems – Part 1: Pipeline Systems for Compressed Medical Gases and Vacuum*.
- NFPA 99, *Health Care Facilities Code*, applicable adopted edition.
- WHO technical guidance on oxygen systems and oxygen-source options for health facilities.
- G. D. Kunders, *Hospitals: Facilities Planning and Management*.
- B. M. Sakharkar, *Principles of Hospital Administration and Planning*.

Next Lecture: Hospital Communication and Information Systems

Thank You

Questions and Discussion

Hospital Engineering and Management – UBM702